Two Kinect v2 Cannot Stream Simultaneously

USB isochronous bandwidth problem — E-Trike

2026-08-24 · Jetson AGX Orin · ROS 2 Humble · libfreenect2 0.2.0

The Symptom

Each Kinect v2 works **individually** at 30 Hz (CUDA) / 5 Hz (CPU). But when both are active at the same time, exactly **one** of them streams and the **other** fails to start and crashes:

- Front-only: front streams, rear never connects.
- Rear-only: rear streams, front fails.
- Both launched: the camera that starts **second** always fails.

The failing node prints:

```
[protocol::UsbControl] failed to set ir interface state! LIBUSB_ERROR_OTHER Other
error.
[kinect_front]: opened serial=500076343042 but failed to start streaming
[kinect2_node_exec]: process has died [pid ..., exit code -11, ...]
```

and the kernel logs:

```
usb 2-3.3.1: Not enough bandwidth for new device state.
usb 2-3.3.1: Not enough bandwidth for altsetting 1
```

The underlying libusb error is:

```
libusb: error [op_set_interface] set interface failed, errno=28
```

errno=28 = ENOSPC — the USB controller refused the isochronous interface configuration.

What “Not enough bandwidth for altsetting 1” Means

The Kinect v2 IR interface (`:1.1`) has two alternate settings:

- **alt setting 0** — no isochronous endpoints (control only)
- **alt setting 1** — enables the isochronous endpoint (endpoint `0x84`, `wMaxPacketSize=0x400`, `bInterval=1`) that carries the raw IR/depth stream

`startStreams()` calls `libusb_set_interface_alt_setting(handle, 1, 1)` to switch on the isochronous stream. On Linux, the xHCI driver must **reserve isochronous bandwidth** on the controller for that endpoint. When the budget is already consumed, the kernel returns `ENOSPC` and the alt-setting switch fails — so libfreenect2 reports `LIBUSB_ERROR_OTHER` on `setIrInterfaceState`.

This is a real USB bandwidth reservation failure — **not a ROS bug, not a libfreenect2 bug, not a CUDA issue** (reproduced with the CPU pipeline too).

The USB Topology on Our Jetson (Root Cause)

```
tegra-xusb controller (Bus 02, SuperSpeed 10000M)
  └── usb2-port3 (root port 3)
        └── 4-port hub (2-3)
              ├── Port 2 ── hub/1p ── Kinect v2 #2 (2-3.2.1, serial 500396543042,
REAR)
              └── Port 3 ── hub/1p ── Kinect v2 #1 (2-3.3.1, serial 500076343042,
FRONT)
```

- Both Kinects are behind the **same 4-port USB hub** on the **same root port 3**.
- A single hub's upstream link / the xHCI's per-root-port isochronous budget cannot hold two full Kinect v2 streams (180 MB/s each: color 1920×1080@30 + IR/depth 512×424@30).
- The **first** camera to start consumes the budget; the **second** cannot reserve its IR interface → fails with `ENOSPC` and segfaults (a libfreenect2 cleanup bug on the failed start).

Evidence

Test	Result
Front alone (CUDA)	✅ 30 Hz
Rear alone (CUDA)	✅ 30 Hz
Front alone (CPU)	✅ 5 Hz
Rear alone (CPU)	✅ 6 Hz
Both, CUDA	❌ one fails `ir interface state` (ENOSPC) + segfault
Both, CPU	❌ same failure (not CUDA-related)
Both, staggered start	❌ same failure
Both, single libfreenect2 context/process	❌ same failure on the first `start()`
`usbfs_memory_mb` 16 → 64	fixed earlier `LIBUSB_ERROR_NO_MEM` did NOT fix bandwidth
Transfer-pool tuning	❌ did not fix
After the failed start	the whole USB bus can wedge → needs a Jetson reboot

Also: even in a single `Freenect2` context opening both devices succeeds, but `start()` on the first device already fails — proving the bandwidth reservation is the hard constraint, not enumeration or multi-process handling.

Why the Hardware Supports It, and What to Do

The AGX Orin's `tegra-xusb` has **4 SuperSpeed root ports** (`usb2-port1` through `usb2-port4`). The hardware is capable of two Kinects — but each needs its own root port so the xHCI controller can allocate independent isochronous bandwidth.

The fix is physical: move one Kinect to a different USB3 connector that routes to a **different root port** than the shared 4-port hub. On the AGX Orin developer kit that is typically the **USB-C port** (separate root port) or a USB-A port that is not behind the same hub.

Success criterion: after re-wiring, `lsusb -t` must show the two `02d8` sensors under **different top-level branches** of Bus 02, e.g.:

```
Bus 02 Port 1: tegra-xusb/4p
   ├── Port 1: ...  Kinect v2 A (direct root port)
   └── Port 3: ...  4-port hub
                      └── Port 2: ... Kinect v2 B
```

NOT both under the same `hub/4p`.

Reference

This matches libfreenect2′s documented multi-Kinect requirement: separate USB 3.0 host controllers (or here, separate root ports), because each Kinect v2 uses 1/3 of a SuperSpeed controller’s isochronous budget and two exceed a single hub/root-port budget.

Issues with the identical error: OpenKinect/libfreenect2 971, 615, 500, 97 — “Not enough bandwidth”.

Generated by Typst. Companion: docs/sensors/kinect2/problem-dual-kinect-bandwidth.md